VIT BHOPAL • CSE4001 CURRICULUM SYNC & FULL-STACK TRACK

Web Development Master Execution Plan

Start Date: Nov 13, 2026 (CSS Part 5)
Target Completion: Mar 31, 2027
Weekly Guardrail: 0 Web Dev on Mondays

> **Schedule Calibration Rules:** Begins **Friday, November 13, 2026** directly with **Chapter 11 (CSS - Part 5)**. Because Mondays run from 08:30 AM to 07:30 PM, **every single Monday is strictly excluded** from Web Development coding or study—all day counts below reflect active Tuesday–Sunday working days. Locked Zero-Web Dev buffers protect your 9.25+ CGPA during **CAT-1 (Dec 13–23)**, **Mid-Sem Catchup (Jan 27–31)**, **CAT-2 (Feb 03–13)**, and **TEE Finals (Mar 11–31)**.

Phase 1: Pre-CAT-1 Sync — CSS3, Responsive Design & Bootstrap (Nov 13 – Dec 12, 2026)

Covers **CSE4001 Unit 2 (HTML5 & CSS3 Basics, Selectors, Box Model, Layouts, Media)** and Lab Tasks 1–3 prior to CAT-1.

Calendar Window	Active Days (Mondays Excluded)	Chapters & Topics	Technical Deliverables & CSE4001 Alignment
Nov 13 – Nov 18	**5 Active Days** (Excl. Mon Nov 16)	**11. CSS - Part 5 & 12. CSS - Part 6**	Flexbox containers, CSS Grid tracks, alignment, transitions, 2D/3D transforms, media queries, and responsive breakpoints (Unit 2).
Nov 19 – Nov 22	**4 Active Days** (Tue–Sun)	**13. Mini Project (CSS)**	Build responsive photography/landing page layout, online Periodic Table layout (Lab Task 3), and image map styling (Lab Task 2).
Nov 24 – Nov 29	**6 Active Days** (Excl. Mon Nov 23)	**14. Bootstrap**	12-column grid system, responsive utilities, navigation bars, modals, forms, cards, and spacing helpers.
Dec 01 – Dec 12	**11 Active Days** (Excl. Mon Nov 30, Dec 7)	**15. Project (CSS)**	Full multi-page responsive web UI clone combining custom CSS3 and Bootstrap components; consolidates 100% of CSE4001 Unit 2.
Dec 13 – Dec 23	**11 Days**	**CAT-1 GPA Guard Buffer & Exams**	**Zero Web Dev Tasks: Dec 13–14 Pre-Exam Buffer + Dec 15–23 CAT-1 Exams. 100% university theory revision across all courses.**
Dec 24 – Dec 25	**2 Days**	**Post-CAT-1 Rest & Christmas**	**Complete physical rest, sleep recovery, and zero screen time before Unit 3 begins.**

Phase 2: Pre-CAT-2 Sync — JavaScript, DOM, Async APIs & Git (Dec 26, 2026 – Feb 02, 2027)

Covers **CSE4001 Unit 3 (JavaScript, DOM, Events, Regex) & Unit 4 (JSON, AJAX basics)**, completing Lab Tasks 4–7.

Calendar Window	Active Days (Mondays Excluded)	Chapters & Topics	Technical Deliverables & CSE4001 Alignment
Dec 26 – Jan 02	**7 Active Days** (Excl. Mon Dec 28)	**16–19. JavaScript (Parts 1, 2, 3 & 4)**	Variables, data types, operators, conditionals, loops, string methods, array manipulation, and nested loops (CSE4001 Unit 3).
Jan 03 – Jan 09	**6 Active Days** (Excl. Mon Jan 04)	**20–23. JavaScript (Parts 5, 6, 7 & 8)**	Object literals, lexical scope, closures, `this` keyword, arrow functions, `map`/`filter`/`reduce`, spread/rest, and destructuring.
Jan 10 – Jan 16	**6 Active Days** (Excl. Mon Jan 11)	**24–25. JavaScript (Parts 9 & 10)**	DOM selection, manipulation, event listeners, event bubbling, form events, Tic-Tac-Toe logic (Lab Task 5), and Matrix transpose (Lab 6).
Jan 17 – Jan 21	**4 Active Days** (Excl. Mon Jan 18)	**26. JavaScript (Mini-Project)**	Interactive DOM-driven game (Simon Says / Tic-Tac-Toe) and validated Hospital Registration Form (CSE4001 Lab Task 7).
Jan 22 – Jan 26	**4 Active Days** (Excl. Mon Jan 25)	**27–29. JavaScript (Parts 11, 12 & 13)**	Call stack, Promises, `async/await`, Fetch API, Axios, JSON parsing, AJAX requests, and Regular Expression ISBN validator (Lab Task 4).
Jan 27 – Jan 31	**5 Days**	**Dedicated No-Web Dev Buffer**	**Zero Web Dev Tasks: 5 full days reserved for clearing Digital Assignments, Lab records, and mid-semester college coursework.**
Feb 02	**1 Active Day** (Excl. Mon Feb 01)	**30. Terminal & 31. Git & Github**	CLI navigation, environment variables, `git init`, branching, merge workflows, and pushing Unit 2–4 lab repositories to GitHub.
Feb 03 – Feb 13	**11 Days**	**CAT-2 GPA Guard Buffer & Exams**	**Zero Web Dev Tasks: Feb 03–04 Pre-Exam Buffer + Feb 05–13 CAT-2 Exams. 100% focus on Units 3 & 4 across all subjects.**

Phase 3: Pre-TEE Backend & Databases — Node, Express, SQL & MongoDB (Feb 14 – Mar 10, 2027)

Covers **CSE4001 Unit 5 (Server-side architecture, Form Handling, Sessions, Cookies & Database Integration)** plus full-stack backend foundations.

Calendar Window	Active Days (Mondays Excluded)	Chapters & Topics	Technical Deliverables & Syllabus Alignment
Feb 14 – Feb 19	**5 Active Days** (Excl. Mon Feb 15)	**32–35. Node.js, Express, EJS & Misc**	Node runtime, `require`/`import`, Express server routing, dynamic EJS templates, partials, GET vs POST parsing, and HTTP status codes.
Feb 20 – Feb 25	**5 Active Days** (Excl. Mon Feb 22)	**36–39. REST APIs, SQL & Node-SQL**	7 RESTful CRUD routes, `method-override`, relational SQL schemas, primary/foreign keys, joins, and `mysql2` connection pools (Unit 5 DB).

Calendar Window	Active Days (Mondays Excluded)	Chapters & Topics	Technical Deliverables & Syllabus Alignment
Feb 26 – Mar 03	**5 Active Days** (Excl. Mon Mar 01)	**40–43. MongoDB & Mongoose**	MongoDB installation, `mongosh`, BSON document collections, Mongoose schemas, model validations, and Express route integration.
Mar 04 – Mar 10	**6 Active Days** (Excl. Mon Mar 08)	**44–54. Full-Stack Project (Phases 1 & 2) + Middleware & Auth**	Listing CRUD routes, custom middleware (`next()`), centralized `ExpressError` handling, DB relationships (`populate()`), Joi validations, sessions, and Passport.js auth.

Phase 4: TEE Window & Modern React Ecosystem Finale (Mar 11 – Mar 31, 2027)

Calibrated around your **Term End Examinations (Mar 11–31)**. Theory exam days carry zero web development tasks; non-exam gap days and weekend dawn blocks finish your React, Tailwind, and Redux Toolkit modules cleanly by March 31.

Calendar Window	Active Days (Mondays & Exam Days Excluded)	Chapters & Topics	Final Capstone & Modern Framework Deliverables
Mar 11 – Mar 31	**Exam Guard**	**TEE Final Exam Immersion**	**Zero Web Dev on Exam Days & Mondays (Mar 15, 22, 29): University finals take 100% priority. Use non-exam study gaps (1–1.5 hrs/day) for Chapters 55–68 below.**
Mar 11 – Mar 17	**4 Gap Days** (Excl. Mon Mar 15)	**55–58. Project Phase 3 (MVC, Cloudinary, Maps & Deploy)**	MVC controller refactoring, Multer & Cloudinary image uploads, Mapbox geocoding, Helmet security headers, and cloud deployment.
Mar 18 – Mar 25	**5 Gap Days** (Excl. Mon Mar 22)	**59–65. React (Parts 1, 2, 3, 4, 5, 6 & Misc)**	Vite build setup, JSX, props, `useState`, complex object/array state, controlled forms, `useEffect` lifecycles, and React Router DOM v6.
Mar 26 – Mar 31	**4 Gap Days** (Excl. Mon Mar 29)	**66–69. React Capstone, Tailwind CSS, Redux Toolkit & Certificate**	Full-stack MERN capstone integration, Tailwind CSS responsive utility layouts, Redux Toolkit global store/slices, and course completion certificate.
Mar 31, 2027	**Target Hit**	**Full-Stack Web Development Complete**	**100% of Chapters 11–69 finished by March end while respecting every Monday blackout and exam buffer.**